Functional Annotation Report: *lcdH* (L-carnitine dehydrogenase, UniProt Q88R32)

Organism: *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) **Gene:** *lcdH* (ordered locus PP_0302) **Protein:** L-carnitine dehydrogenase (CDH / L-CDH), EC 1.1.1.108 **UniProt:** Q88R32 · HAMAP rule MF_02129

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0. Gene identity verification

The target was verified against all provided UniProt criteria before research:

- **Gene symbol / function match:** The symbol *lcdH* denotes **l**-**c**arnitine **deh**ydrogenase, consistent with the UniProt description "L-carnitine dehydrogenase, EC 1.1.1.108."
- **Organism match:** Primary biochemical literature characterizes carnitine dehydrogenase from *Pseudomonas putida* (strain IFP 206) — the same species as the target strain KT2440 — with the identical EC number 1.1.1.108 (Goulas 1988, P 3058208). KT2440-specific pathway context is corroborated by studies in the closely related *P. aeruginosa* and by classic *P. putida* physiology work (Kleber 1978, P 565193).
- **Family / domain match:** Literature and UniProt agree that the enzyme belongs to the 3-hydroxyacyl-CoA dehydrogenase family and carries an N-terminal NAD-binding Rossmann domain (IPR006176) and C-terminal helical catalytic domain (IPR006108), matching the provided domain list.

Conclusion: The gene symbol is unambiguous for this protein and the literature is directly on-target. No conflicting-gene problem was encountered. Research proceeded with high confidence.

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1. Summary (answer to the research question)

lcdH encodes **L-carnitine dehydrogenase (EC 1.1.1.108)**, a soluble, cytoplasmic, NAD^+-dependent oxidoreductase that catalyzes the **reversible oxidation of L-carnitine to 3-dehydrocarnitine**:

> **L-carnitine** + **NAD^+** ⇌ **3-dehydrocarnitine** + **NADH** + **H^+**

The enzyme is **strictly specific for the L-(−)-enantiomer of carnitine and for NAD^+**. It performs the **committed first step of aerobic L-carnitine catabolism**, enabling *P. putida* to use L-carnitine as a sole source of carbon, nitrogen, and energy. Its product feeds a downstream pathway (3-dehydrocarnitine → glycine betaine → dimethylglycine → sarcosine → glycine). Structurally it is a homodimer of the 3-hydroxyacyl-CoA dehydrogenase (HAD) superfamily fold, acting in the cytoplasm.

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2. Key findings with evidence

2.1 Catalyzed reaction and cofactor

Carnitine dehydrogenase (carnitine:NAD^+ oxidoreductase, EC 1.1.1.108) purified from *P. putida* IFP 206 "catalyzes the oxidation of L-carnitine to 3-dehydrocarnitine" using NAD^+ (Goulas 1988, P 3058208).

The reaction is reversible; the enzyme has distinct pH optima for the two directions (pH 9.0 for oxidation, pH 7.0 for the reduction of 3-dehydrocarnitine back to L-carnitine), a hallmark of a thermodynamically balanced dehydrogenase.

Mechanistically, the enzyme oxidizes the **C3 secondary hydroxyl** of carnitine to a **C3 keto group**, exactly analogous to how 3-hydroxyacyl-CoA dehydrogenases oxidize the 3-hydroxyl of their substrates — consistent with its assignment to that structural family (UniProt Q88R32; IPR026578 L-carnitine_dehydrogenase signature).

2.2 Substrate and stereospecificity

The enzyme "is specific for L-carnitine and NAD^+" (Goulas 1988, 3058208): it does not oxidize D-carnitine and does not substitute $NADP^+$ for NAD^+. Km values for both substrates were determined. This stereospecificity is mirrored physiologically — *P. putida* and *Acinetobacter* grow on L-carnitine but not on D-carnitine as a sole carbon source unless L-carnitine is co-supplied (Kleber et al. 1977, 849100; Kleber et al. 1978, 565193).

2.3 Physical / biochemical properties

- Molecular mass ≈ **62 kDa**, a **homodimer of two identical subunits** (~31 kDa each) (Goulas 1988, 3058208).
- Isoelectric point **4.7**; temperature optimum **30 °C**; purified **72-fold to homogeneity**.

2.4 Pathway role — committed first step of carnitine catabolism

"Some, especially *Pseudomonas* species, assimilate L-(−)-carnitine as sole source of carbon and nitrogen. **The first catabolic step is catalysed by the L-(−)-carnitine dehydrogenase**" (Kleber 1997 review, 9037756).

The product, 3-dehydrocarnitine, "is in turn metabolized to glycine betaine (GB), an intermediate metabolite in the catabolism of carnitine to glycine" (Wargo & Hogan 2009, 19406895).

Downstream, glycine betaine is demethylated stepwise to dimethylglycine → sarcosine → glycine by the GB-catabolic genes (gbcAB, dgcAB, soxBDAG) (Wargo et al. 2008, 17951379).

The deacetylation of 3-dehydrocarnitine to glycine betaine is proposed to involve a 3-ketoacid CoA-transferase (PA1999-PA2000 in *P. aeruginosa*) (Wargo & Hogan 2009, P 19406895).

Full catabolic route initiated by lcdH: L-carnitine → **[lcdH]** → 3-dehydrocarnitine → glycine betaine → dimethylglycine → sarcosine → glycine.

2.5 Regulation

Expression is **substrate-inducible**: "L-Carnitine induced the carnitine dehydrogenase," and γ-butyrobetaine also induces it; synthesis "was repressed by glycinebetaine, glucose and long-chain fatty acids" (Kleber et al. 1978, P 565193). This links enzyme production to carnitine availability and to catabolite/pathway-product repression, so the cell makes the enzyme only when L-carnitine is present and preferred carbon sources are absent.

2.6 Localization

The enzyme was recovered from **soluble crude extract** and purified without membrane solubilization (Goulas 1988, P 3058208 ; activities measured in crude extract, Kleber 1978, P 565193), indicating a **cytoplasmic** location. This is where imported L-carnitine is oxidized. No signal/transmembrane features are indicated for this HAD-family soluble dehydrogenase.

2.7 Structural inference

UniProt Q88R32 places the protein in the **3-hydroxyacyl-CoA dehydrogenase family** with: - **N-terminal NAD-binding Rossmann domain** (IPR006176, 3-OHacyl-CoA_DH_NAD-bd) — binds NAD^+. - **C-terminal α-helical catalytic/dimerization domain** (IPR006108, 3HC_DH_C; 6-phosphogluconate-dehydrogenase-like fold IPR008927/IPR013328) — forms the dimer interface and completes the active site. - **L-carnitine dehydrogenase signature** (IPR026578).

The biochemically observed homodimer (Goulas 1988) matches this two-domain HAD fold, in which each subunit contributes a Rossmann NAD-binding module and a helical domain, and dimerization is required for a functional active site.

2.8 Sequence-level active-site evidence (bioinformatic)

Direct analysis of the 321-residue Q88R32 sequence (UniProt REST) confirms the diagnostic motifs of the family: - **NAD-binding Rossmann fingerprint** `G-S-G-V-I-G` at residues **14–19** — exactly the UniProt-annotated NAD binding site (14–19). This glycine-rich βαβ loop is the classic dinucleotide-binding signature. - **Substrate-binding motif** `SSTSG` at residues **117–121**, a conserved segment of the 3-hydroxyacyl-CoA dehydrogenase substrate pocket. - **Family-diagnostic catalytic motif** `GHPFNP` at residues **140–145**, containing catalytic **His141** — the general-base histidine that deprotonates the C3-hydroxyl of L-carnitine, enabling hydride transfer to NAD^+ to form the 3-keto product. This is the same catalytic logic used by 3-hydroxyacyl-CoA dehydrogenases.

UniProt/HAMAP (rule MF_02129) annotations independently confirm the manual findings: FUNCTION "Catalyzes the NAD(+)-dependent oxidation of L-carnitine to 3-dehydrocarnitine"; CATALYTIC "carnitine + NAD^+ = 3-dehydrocarnitine + NADH + H^+"; PATHWAY "Amine and polyamine metabolism; carnitine metabolism"; SUBUNIT "Homodimer"; SIMILARITY "3-hydroxyacyl-CoA dehydrogenase family, L-carnitine dehydrogenase subfamily." Database cross-references: KEGG `ppu:PP_0302`, BioCyc `PPUT160488:G1G01-334-MONOMER`, STRING `160488.PP_0302`.

2.9 Genomic context — a dedicated carnitine catabolic cluster (KT2440-specific)

KEGG genome annotation shows that *lcdH* (PP_0302) lies within a compact, co-oriented carnitine-catabolism gene cluster in KT2440, giving direct strain-specific support for the pathway model:

Locus	Product	KO / EC	Role in pathway
PP_0304	Carnitine/glycine betaine ABC-transporter substrate-binding protein	K02002	**Import** of L-carnitine into the cytoplasm
PP_0302 (lcdH)	L-carnitine (carnitine 3-)dehydrogenase	K17735 / EC 1.1.1.108	**Step 1:** L-carnitine → 3-dehydrocarnitine (NAD^+)
PP_0303	Dehydrocarnitine cleavage enzyme (3-dehydrocarnitine:acetyl-CoA trimethylamine transferase)	K27837 / EC 2.3.1.317	**Step 2:** 3-dehydrocarnitine → glycine betaine (removes acetyl unit as acetyl-CoA)
PP_0301	Betainyl-CoA thioesterase	K27492 / EC 3.1.2.33	Accessory CoA-ester hydrolysis
PP_0305	CdhR, AraC-family activator	K17736	Carnitine-catabolism transcriptional **activator**
PP_0298	AraC-family, glycine betaine-responsive activator	K21826	Downstream/betaine-responsive regulation

This gene order (transporter → dehydrogenase → cleavage enzyme → regulator) confirms that L-carnitine is first **imported** by the PP_0304 ABC transporter, then oxidized **in the cytoplasm** by lcdH, after which the adjacent PP_0303 cleavage enzyme converts the 3-dehydrocarnitine product to **glycine betaine**. The clustered activator **CdhR (PP_0305)** provides the molecular basis for the substrate-inducibility of the enzyme observed biochemically (Kleber 1978, P 565193).

Notably, this KT2440 cluster identifies the downstream betaine-forming step precisely (EC 2.3.1.317), refining the earlier "3-ketoacid CoA-transferase" inference drawn from *P. aeruginosa*.

3. Supported and refuted hypotheses

Hypothesis	Verdict	Basis
lcdH encodes an NAD^+-dependent L-carnitine dehydrogenase (EC 1.1.1.108) oxidizing L-carnitine → 3-dehydrocarnitine	**Supported**	Goulas 1988 (P 3058208); UniProt/HAMAP MF_02129
Enzyme is stereospecific (L-carnitine only) and NAD^+-specific (not $NADP^+$)	**Supported**	Goulas 1988 (P 3058208)
It performs the committed first step of carnitine catabolism enabling growth on carnitine as C/N source	**Supported**	Kleber 1997 (P 9037756); Wargo & Hogan 2009 (P 19406895)
Product 3-dehydrocarnitine is routed to glycine betaine → glycine	**Supported**	Wargo & Hogan 2009 (P 19406895); Wargo 2008 (P 17951379)
Enzyme is cytoplasmic (soluble)	**Supported (strong inference)**	Soluble purification (P 3058208 P 565193)
Homodimeric HAD-family fold with Rossmann NAD-binding domain	**Supported**	P 3058208 + UniProt domain architecture + sequence motifs (GSGVIG, His141)
lcdH sits in a dedicated carnitine catabolic cluster (importer + dehydrogenase + cleavage enzyme + CdhR activator)	**Supported (KT2440 genomics)**	KEGG PP_0298–PP_0305
Enzyme uses $NADP^+$ or acts on D-carnitine	**Refuted**	Goulas 1988 (P 3058208):

Hypothesis	Verdict	Basis
		specific for L-carnitine and NAD^+

4. Limitations and future directions

- **Strain provenance:** The definitive biochemical characterization (Goulas 1988) used *P. putida* IFP 206, not KT2440 directly; genetic dissection of the pathway was done in *P. aeruginosa* PA14. KT2440 (PP_0302) function is assigned by sequence/HAMAP orthology, which is well supported but not yet KT2440-specific enzymology.
- **No experimental 3D structure** of this specific enzyme was located; the fold description rests on family/domain inference. AlphaFold modeling and superposition onto 3-hydroxyacyl-CoA dehydrogenase structures could confirm the active-site geometry and identify catalytic His/Asn residues.
- **Downstream deacetylation step** (3-dehydrocarnitine → glycine betaine) mechanism is inferred (3-ketoacid CoA-transferase) but not fully enzymatically characterized in *P. putida*.
- **Future work:** (i) KT2440 ΔPP_0302 knockout growth assays on L-carnitine; (ii) recombinant Q88R32 kinetics (Km, kcat for L-carnitine/NAD^+); (iii) crystal/cryo-EM structure with NAD^+/substrate to define catalytic residues; (iv) transcriptional mapping of the PP_0302 gene cluster and its regulator.

References (PMID)

- **3058208** — Goulas P. (1988) Purification and properties of carnitine dehydrogenase from *Pseudomonas putida*. *Biochim Biophys Acta*.
- **9037756** — Kleber H-P. (1997) Bacterial carnitine metabolism. *FEMS Microbiol Lett*.
- **19406895** — Wargo MJ, Hogan DA. (2009) Identification of genes required for *Pseudomonas aeruginosa* carnitine catabolism. *Microbiology*.
- **17951379** — Wargo MJ, Szwergold BS, Hogan DA. (2008) Two gene clusters and a regulator required for *P. aeruginosa* glycine betaine catabolism. *J Bacteriol*.

- **565193** — Kleber H-P, Seim H, Aurich H, Strack E. (1978) Interrelationships between carnitine metabolism and fatty acid assimilation in *Pseudomonas putida*.
- **849100** — Kleber H-P, Seim H, Aurich H, Strack E. (1977) Utilization of trimethylammonium compounds by *Acinetobacter calcoaceticus*.
- **30670548** — Bazire P, et al. (2019) Characterization of L-carnitine metabolism in *Sinorhizobium meliloti. J Bacteriol.*
